



Scientists using PS3 to model black holes, drug molecules

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BY AFP FEBRUARY 13, 2008



PARIS -- Researchers are cannibalising the Sony PlayStation 3 console and other gaming hardware, turning them into low-cost supercomputers to model pharmaceutical molecules and black holes, the weekly New Scientist says.

The PS3 console uses a Cell chip, made by IBM, Sony and Toshiba, that comprises a central processing unit and eight slave processors and can run on the open-access Linux operating system.

The chip is prized by chemists and physicists because the kinds of calculations required to make high-quality graphics for games are similar to those used to simulate reactions between particles, ranging from the molecular to the astronomical.

University of Massachusetts astrophysicist Gaurav Khanna has strung together 16 PS consoles to simulate gravity waves that occur when two black holes collide, the British magazine says.

Another innovation is a graphics processor made by NVIDIA that boosts gaming-image quality in personal computers.

By using the C programming language to run the chip, University of Illinois chemist Todd Martinez found he could run calculations 130 times faster than on an ordinary PC.

His is now calculating the energy of the electrons in 1,000 atoms, which add up to the size of a small protein.

"We can now do the things we were killing ourselves to do," Martinez told New Scientist.

Another newly-harnessed scientific tool taken from the world of gaming is the Wii hand-held remote control, which contains an accelerometer that can sense in which direction it is being moved and how vigorously.

Surgeons have been using the Wii as a gadget to improve their operating technique, and neurologists have turned to it to measure movement deficiencies in patients who have had a stroke or have Parkinson's disease.

The report appears in Saturday's issue of New Scientist.

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